



IESA DeepTech Hackathon

Idea Submission Template

EDGEVISION

Team Details

Team Name:

Enter Team Name Here...

SR. NO	ROLE	NAME	ACADEMIC YEAR
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Problem Statement Addressed



Edge-AI Defect Classification for Semiconductor Images

- Design and build an Edge-AI powered defect classification system capable of:
- Detecting and classifying defects in wafer/die images into predefined categories
- Demonstrating strong accuracy with lightweight compute (edge constraints)
- Supporting real-time, high-volume inspection workflows by being portable to edge deployment flows (NXP eIQ)

Idea Description – EdgeInspect AI



KEY CONCEPT & APPROACH

Our core idea is to bring smart, real-time AI inspection directly onto NXP edge hardware so defect detection happens at the production line itself instead of in the cloud. We use an AI vision model that not only classifies wafer defects but also measures confidence, learns from uncertain cases, and continuously improves with more data over time.

SOLUTION OVERVIEW

The system automatically inspects wafer images, predicts defect types, logs results, and flags low-confidence samples for future learning. It works offline, analyzes entire batches, calculates yield, and builds a growing digital quality record. By combining edge AI, automation, and self-learning, the solution reduces manual errors, speeds up inspection, and makes wafer quality control smarter, scalable, and more reliable for industry.

Proposed Solution - Describe your Idea/Solution/Prototype



CORE APPROACH

- AI runs directly on **NXP i.MX RT (Edge device)** — fast, real-time inspection on the factory line
- No dependence on cloud or continuous internet

Smart Wafer Inspection System

Real-Time Defect Detection

The system analyzes wafer images in real-time to identify defects.

Uncertain Case Handling

Uncertain cases are flagged for future learning and improvement.

Yield Estimation

The system estimates the overall yield of wafer batches.

Continuous Improvement

The system learns from new data to improve its accuracy over time.



Defect Prediction and Confidence

The system predicts defect types and provides a confidence score for each.

Batch Inspection and Digital Recording

The system inspects entire batches and records results digitally.

Offline Operation

The system can operate without an internet connection.

Proposed Solution - Describe your Idea/Solution/Prototype



Methodology

- Standardize wafer images → AI model predicts defect + confidence.
- Works for single images and full batches.
- Batch results are automatically aggregated and yield is computed.

Pipeline Logic

- High-confidence results are logged with timestamp.
- Low-confidence images are stored in a Self-Learning folder

Technology

- TensorFlow/Keras model with **MobileNetV3Small** backbone.
- Grayscale images converted to 3-channel format for the model.
- Designed for conversion to **TensorFlow Lite** for NXP deployment.

Implementation Strategy

- Structured folders for inputs, results, and learning data.
- Final merged pipeline prepared for execution on **NXP i.MX RT** hardware.

Innovation and Uniqueness



KEY INNOVATION

- **AI moved to the factory floor (edge device), not cloud**
- **Confidence-aware predictions instead of blind labels**
- **Automatic self-learning loop from real production data**
- **One unified pipeline for detection, logging, and batch analysis**
- **System keeps getting better as more images are added**

COMPETITIVE ADVANTAGE

- **Edge + Offline:** Works on-site without internet.
- **Confidence-based AI:** Knows when it is unsure.
- **Self-learning system:** Automatically collects weak cases to improve.
- **One unified pipeline:** Detection, logging, and batch analysis together.
- **Factory-ready design:** Built for real industrial workflow

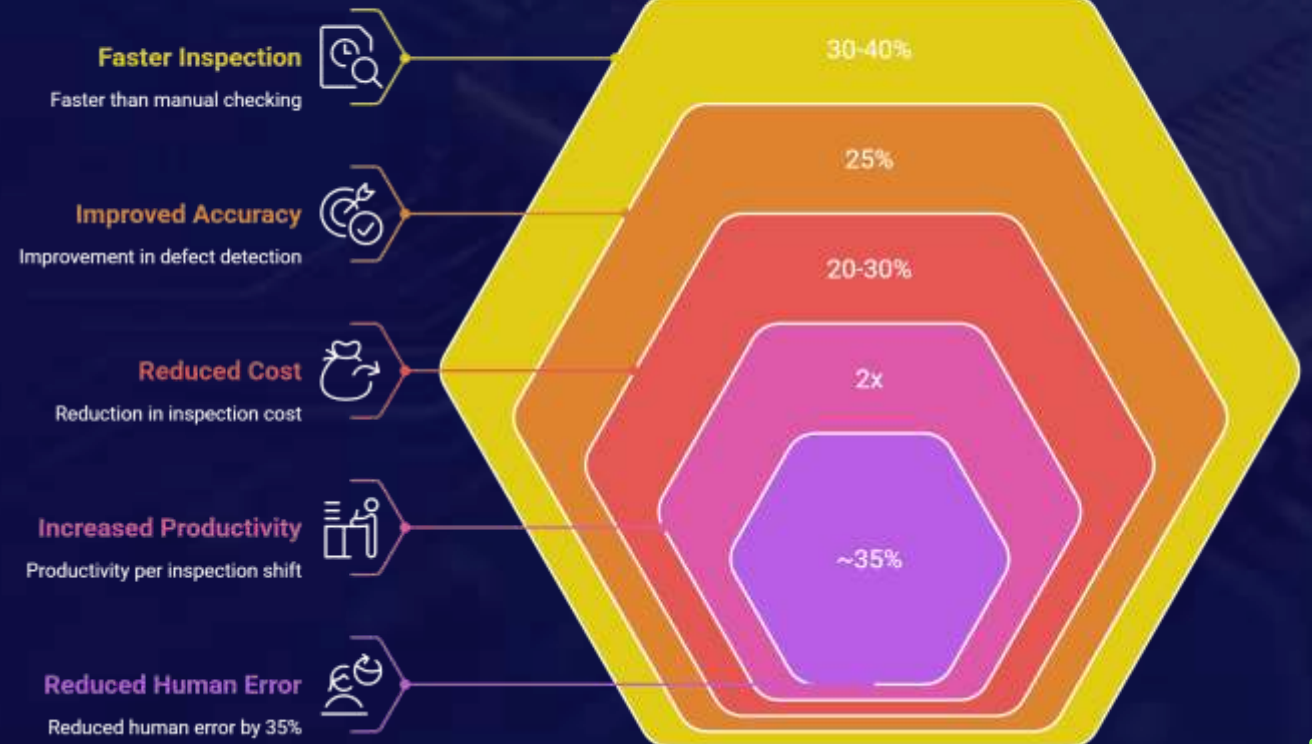
Impact and Benefits



Primary Impact and Quantifiable Outcomes

It improves yield and efficiency by automating inspection with a self-learning, edge-based AI system that works without internet.

Automated Inspection System Benefits



Dataset Structure and Model Performance



Total Images (Current): ~1,100 images

Number of Classes: 8

(6 Defects + Clean + Other)

Class List:

Bridge, Crack, CMP, LER, Open, Vias, Clean, Other

Class Balance Plan:

Minimum **120–150 images per class** (where possible)

Train / Val / Test Split:

70% / 20% / 10%

Image Type:

Grayscale images (converted to 3-channel for model input)

Labeling Method / Source:

Manually labeled + Public wafer defect dataset

Architecture: MobileNetV3Small
(Lightweight CNN)

Training approach: Transfer Learning

Input size: $224 \times 224 \times 3$

Model size: 1.9 MB

Framework: TensorFlow

Metrics on test split

Accuracy: ~75%

Val Accuracy :~ 80%

Precision: ~0.74

Recall: ~0.76

GitHub & Video Link



GitHub Repository

<https://github.com/MithunSrinivas28/wafer-defect-ai>



Prototype / Simulation Video

https://drive.google.com/file/d/1Ki0E7Ki41A_NLVaYZ_79rIJnnP1FOWBb/view?usp=drive_link



Research Background & Methodology

Based on computer vision using CNNs, proven for visual defect detection in manufacturing.
Uses a lightweight edge AI pipeline that predicts defects, checks confidence, and supports continual learning.